

Matched Pair Design Results - Replication of Christensen & Timmins (2022)

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Replication Summary

- Choice set size (Table 5): minorities receive fewer recommended properties in total and per appointment; advertised-property availability differences are small and not significant.
- Segregation (Tables 6–7): minority testers, especially African American, are steered to lower white-share neighborhoods, strongest in high- and middle-income strata.
- School quality and safety (Tables 8A, 10A): effects are mostly null; the only consistent signal is modestly lower SEDA elementary scores for African American testers.
- ACS neighborhood characteristics (Tables 8B, 10B): lower high-skill and college shares and higher single-parent shares for minorities, especially African American and Hispanic; poverty and ownership-rate effects are mostly null.
- Pollution (Tables 9A–9B): full sample shows no significant Superfund, RSEI, or PM2.5 differences; mothers show higher Superfund counts and PM2.5, but RSEI remains null.
- Median income (Table 12): modest negative effects only for African American families and mothers; otherwise null.

Original Abstract from Christensen & Timmins (2022)

Growing evidence indicates that neighborhoods affect human capital accumulation, raising concern that the exclusionary effects of housing discrimination could contribute to persistent inequality in the United States. Using data from HUD’s most recent Housing Discrimination Study and microlevel data on neighborhood attributes in 28 US cities, we find that minorities are steered toward neighborhoods with less economic opportunity and greater exposures to crime and pollution. Holding preferences and income constant, discriminatory steering alone can explain a disproportionate number of minority households found in high-poverty neighborhoods in the United States and the higher exposure of African American mothers to toxic pollutants.

Largest Differences vs. Christensen & Timmins (2022)

- Quantity vs. quality flips: we find significant reductions in the number of recommended properties (Table 5), while the paper emphasizes no meaningful quantity effects.

- School and crime channel weakens: our school-quality and assaults results are mostly null, unlike the strong negative school and higher crime effects reported in the paper.
- Pollution effects attenuate: full-sample pollution estimates are not significant; for mothers, only Superfund and PM2.5 show positive differences, while RSEI does not.
- Mother-specific effects are smaller and less pervasive than the paper's reported magnitudes.

Table 1: Discriminatory Steering and Availability of Advertised Properties *Table 5, C&T 2022*

	Dependent Variable			
	Total Recommended Properties	Ad Property Ever Available	Recommended Properties per Appointment	Ad Property Available per Appointment
Racial Minority	-0.3099*** (0.0822) [-0.4710,-0.1488]	-0.0128 (0.0098) [-0.0320, 0.0065]	-0.1864*** (0.0709) [-0.3254,-0.0474]	0.0037 (0.0096) [-0.0151, 0.0225]
African American	-0.4569*** (0.1322) [-0.7161,-0.1977]	-0.0079 (0.0159) [-0.0390, 0.0233]	-0.2996*** (0.1150) [-0.5250,-0.0742]	0.0129 (0.0156) [-0.0175, 0.0434]
Hispanic	-0.0925 (0.1436) [-0.3740, 0.1890]	-0.0126 (0.0167) [-0.0453, 0.0202]	-0.0587 (0.1279) [-0.3095, 0.1921]	-0.0024 (0.0163) [-0.0344, 0.0296]
Asian	-0.3648*** (0.1197) [-0.5995,-0.1301]	-0.0180 (0.0162) [-0.0497, 0.0137]	-0.1903* (0.1063) [-0.3987, 0.0180]	-0.0001 (0.0160) [-0.0314, 0.0312]
Observations	7262	7262	8291	8291
Adjusted R ² (Minority)	0.1613	0.3881	0.1132	0.3099
Adjusted R ² (Category)	0.1618	0.3878	0.1133	0.3097
Number of Trials	3631	3631	3631	3631

Cluster-robust standard errors in parentheses. Clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ Table 2: Discriminatory Steering and Neighborhood Racial Composition *Table 6, C&T 2022*

	Dependent variable
	White Household Share
Racial Minority	-0.0106*** (0.0032) [-0.0169, -0.0044]
African American	-0.0194*** (0.0049) [-0.0291, -0.0098]
Hispanic	-0.0028 (0.0051) [-0.0129, 0.0073]
Asian	-0.0082* (0.0050) [-0.0180, 0.0016]
Observations	18,889
Adjusted R ² (Minority)	0.7373
Adjusted R ² (Category)	0.7375
Number of Trials	2,702

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Discriminatory Steering and Neighborhood Racial Composition by Income

Table 7, C&T 2022

	Dependent variable		
	High Income	Middle Income	Low Income
Racial Minority	-0.0070* (0.0039) [-0.0145, 0.0006]	-0.0092*** (0.0034) [-0.0159, -0.0026]	-0.0053 (0.0057) [-0.0165, 0.0059]
African American	-0.0205*** (0.0064) [-0.0332, -0.0079]	-0.0151*** (0.0052) [-0.0253, -0.0048]	-0.0073 (0.0089) [-0.0247, 0.0102]
Hispanic	0.0035 (0.0062) [-0.0087, 0.0157]	-0.0029 (0.0055) [-0.0138, 0.0079]	0.0005 (0.0101) [-0.0193, 0.0203]
Asian	-0.0024 (0.0054) [-0.0130, 0.0081]	-0.0088 (0.0061) [-0.0208, 0.0031]	-0.0096 (0.0081) [-0.0254, 0.0063]
Observations	5,225	4,878	5,359
Adjusted R ² (Minority)	0.7533	0.8173	0.8188
Adjusted R ² (Category)	0.7540	0.8173	0.8187
Number of Trials	975	1,051	1,087

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Discriminatory Steering and Neighborhood Effects (Panel A: School Quality and Safety)

Table 8, C&T 2022

	Dependent variable			
	SEDA Elementary	SEDA Middle	Assaults	GreatSchools Elem
Racial Minority	-0.0009 (0.0083) [-0.0173, 0.0154]	-0.0069 (0.0068) [-0.0202, 0.0063]	3.4807 (3.6473) [-3.6761, 10.6375]	-0.0868 (0.0597) [-0.2040, 0.0304]
African American	-0.0226* (0.0134) [-0.0488, 0.0036]	-0.0169 (0.0106) [-0.0378, 0.0039]	5.3515 (5.2389) [-4.9285, 15.6315]	-0.0787 (0.0951) [-0.2654, 0.1080]
Hispanic	-0.0064 (0.0118) [-0.0296, 0.0168]	-0.0057 (0.0120) [-0.0293, 0.0180]	6.1596 (5.0290) [-3.7085, 16.0276]	-0.1540 (0.0982) [-0.3467, 0.0388]
Asian	0.0316** (0.0138) [0.0044, 0.0587]	0.0048 (0.0104) [-0.0156, 0.0253]	-0.6672 (6.8565) [-14.1212, 12.7868]	-0.0406 (0.1003) [-0.2374, 0.1562]
Observations	10,796	11,333	6,612	6,726
Adjusted R ² (Minority)	0.6173	0.7944	0.6532	0.5209
Adjusted R ² (Category)	0.6178	0.7945	0.6532	0.5208
Number of Trials	1,609	1,693	1,044	1,048

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Discriminatory Steering and Neighborhood Effects (Panel B: ACS Characteristics)

Table 8, C&T 2022

	Dependent variable				
	Poverty Rate	High Skill	College	Single Family	Ownership Rate
Racial Minority	0.0011 (0.0013) [-0.0015, 0.0036]	-0.0044** (0.0021) [-0.0085, -0.0002]	-0.0076*** (0.0026) [-0.0127, -0.0026]	0.0033*** (0.0012) [0.0009, 0.0057]	-0.0015 (0.0030) [-0.0074, 0.0044]
African American	0.0008 (0.0018) [-0.0027, 0.0043]	-0.0040 (0.0031) [-0.0101, 0.0020]	-0.0114*** (0.0040) [-0.0192, -0.0036]	0.0055*** (0.0019) [0.0017, 0.0093]	-0.0033 (0.0048) [-0.0126, 0.0061]
Hispanic	0.0015 (0.0021) [-0.0026, 0.0056]	-0.0099*** (0.0033) [-0.0164, -0.0033]	-0.0111*** (0.0041) [-0.0191, -0.0031]	0.0033* (0.0020) [-0.0005, 0.0072]	-0.0048 (0.0051) [-0.0147, 0.0051]
Asian	0.0009 (0.0022) [-0.0033, 0.0052]	0.0001 (0.0039) [-0.0076, 0.0078]	-0.0006 (0.0046) [-0.0096, 0.0084]	0.0008 (0.0020) [-0.0031, 0.0048]	0.0033 (0.0049) [-0.0064, 0.0129]
Observations	18,889	18,889	18,889	18,889	18,889
Adjusted R ² (Minority)	0.5224	0.6587	0.7361	0.5339	0.5583
Adjusted R ² (Category)	0.5224	0.6588	0.7362	0.5340	0.5583
Number of Trials	2,702	2,702	2,702	2,702	2,702

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Discriminatory Steering and Local Pollution Exposures (Panel A: Full Sample)

Table 9, C&T 2022

	Dependent variable		
	Superfund	Toxics (RSEI)	PM2.5
Racial Minority	0.0258 (0.0197) [-0.0128, 0.0644]	-1453.6525 (1374.7067) [-4149.2360, 1241.9310]	-0.0003 (0.0124) [-0.0245, 0.0240]
African American	0.0291 (0.0204) [-0.0110, 0.0692]	-5010.3189 (4082.4454) [-13015.3520, 2994.7141]	0.0083 (0.0182) [-0.0273, 0.0439]
Hispanic	-0.0023 (0.0266) [-0.0545, 0.0498]	512.0068 (713.6044) [-887.2592, 1911.2728]	-0.0328 (0.0207) [-0.0734, 0.0079]
Asian	0.0471 (0.0376) [-0.0266, 0.1208]	575.4138 (511.3807) [-427.3232, 1578.1508]	0.0192 (0.0207) [-0.0214, 0.0599]
Observations	18,897	18,897	18,871
Adjusted R ² (Minority)	0.6571	0.3247	0.9435
Adjusted R ² (Category)	0.6571	0.3247	0.9435
Number of Trials	2,702	2,702	2,701

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Discriminatory Steering and Local Pollution Exposures (Panel B: Mothers)

Table 9, C&T 2022

	Dependent variable		
	Superfund	Toxics (RSEI)	PM2.5
Racial Minority	0.0996** (0.0412) [0.0188, 0.1805]	-1220.6494 (2101.9041) [-5345.3382, 2904.0394]	0.0387* (0.0218) [-0.0041, 0.0816]
African American	0.0704* (0.0408) [-0.0096, 0.1505]	-4091.2189 (4488.2187) [-12898.7120, 4716.2741]	0.0709** (0.0312) [0.0098, 0.1320]
Hispanic	0.0419 (0.0452) [-0.0468, 0.1305]	-257.8142 (1536.6836) [-3273.3380, 2757.7096]	-0.0359 (0.0325) [-0.0997, 0.0278]
Asian	0.1970** (0.0790) [0.0420, 0.3521]	1190.7782 (917.8413) [-610.3553, 2991.9117]	0.0805** (0.0380) [0.0059, 0.1552]
Observations	7,093	7,093	7,089
Adjusted R ² (Minority)	0.6801	0.3439	0.9496
Adjusted R ² (Category)	0.6808	0.3443	0.9497
Number of Trials	993	993	993

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Discriminatory Steering and Neighborhood Effects (Mothers, Panel A)

Table 10, C&T 2022

	Dependent variable			
	SEDA Elementary	SEDA Middle	Assaults	GreatSchools Elem
Racial Minority	0.0013 (0.0131) [-0.0245, 0.0270]	-0.0194* (0.0112) [-0.0413, 0.0026]	1.6264 (3.6276) [-5.5045, 8.7574]	0.0181 (0.1201) [-0.2180, 0.2541]
African American	-0.0151 (0.0208) [-0.0560, 0.0258]	-0.0462** (0.0182) [-0.0820, -0.0103]	5.8079 (5.8319) [-5.6563, 17.2720]	0.0478 (0.1916) [-0.3289, 0.4244]
Hispanic	-0.0052 (0.0193) [-0.0431, 0.0326]	0.0101 (0.0169) [-0.0231, 0.0434]	7.4495 (6.6638) [-5.6501, 20.5492]	-0.1345 (0.1550) [-0.4392, 0.1702]
Asian	0.0368 (0.0252) [-0.0127, 0.0863]	-0.0188 (0.0182) [-0.0545, 0.0169]	-7.3912 (5.9077) [-19.0043, 4.2220]	0.1188 (0.1949) [-0.2643, 0.5019]
Observations	4,132	4,366	2,676	2,696
Adjusted R ² (Minority)	0.6272	0.7965	0.7502	0.4533
Adjusted R ² (Category)	0.6275	0.7970	0.7505	0.4533
Number of Trials	600	627	410	400

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Discriminatory Steering and Neighborhood Effects (Mothers, Panel B)

Table 10, C&T 2022

	Dependent variable				
	Poverty Rate	High Skill	College	Single Family	Ownership Rate
Racial Minority	-0.0003 (0.0019) [-0.0039, 0.0033]	-0.0025 (0.0036) [-0.0095, 0.0046]	-0.0071 (0.0044) [-0.0157, 0.0016]	0.0022 (0.0019) [-0.0016, 0.0060]	-0.0010 (0.0054) [-0.0116, 0.0096]
African American	0.0006 (0.0027) [-0.0046, 0.0059]	-0.0051 (0.0048) [-0.0145, 0.0044]	-0.0168*** (0.0062) [-0.0289, -0.0047]	0.0056* (0.0030) [-0.0004, 0.0115]	-0.0123 (0.0082) [-0.0283, 0.0037]
Hispanic	0.0006 (0.0030) [-0.0053, 0.0066]	-0.0092 (0.0056) [-0.0202, 0.0019]	-0.0107 (0.0073) [-0.0250, 0.0035]	0.0030 (0.0031) [-0.0031, 0.0091]	0.0025 (0.0083) [-0.0138, 0.0188]
Asian	-0.0024 (0.0034) [-0.0091, 0.0043]	0.0079 (0.0069) [-0.0057, 0.0214]	0.0085 (0.0077) [-0.0066, 0.0235]	-0.0028 (0.0034) [-0.0095, 0.0038]	0.0089 (0.0086) [-0.0080, 0.0258]
Observations	7,090	7,090	7,090	7,090	7,090
Adjusted R ² (Minority)	0.5252	0.6388	0.7142	0.5389	0.4751
Adjusted R ² (Category)	0.5252	0.6392	0.7147	0.5391	0.4753
Number of Trials	993	993	993	993	993

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Discriminatory Steering: Median Income in Neighborhood Table 12, C&T 2022

	Dependent variable		
	All Testers	Families	Moms
African American	-0.0110 (0.0089) [-0.0284, 0.0063]	-0.0171* (0.0101) [-0.0369, 0.0027]	-0.0252* (0.0146) [-0.0538, 0.0034]
Hispanic	-0.0115 (0.0094) [-0.0300, 0.0070]	-0.0138 (0.0118) [-0.0370, 0.0094]	-0.0024 (0.0161) [-0.0339, 0.0292]
Asian	0.0013 (0.0099) [-0.0182, 0.0208]	0.0036 (0.0117) [-0.0194, 0.0265]	0.0256 (0.0159) [-0.0056, 0.0568]
Observations	18,889	11,986	7,222
Adjusted R ²	0.6369	0.6359	0.6247
Number of Trials	2,702	2,702	2,702

Cluster-robust standard errors in parentheses; clustered at the trial level. 95% confidence intervals in square brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$